

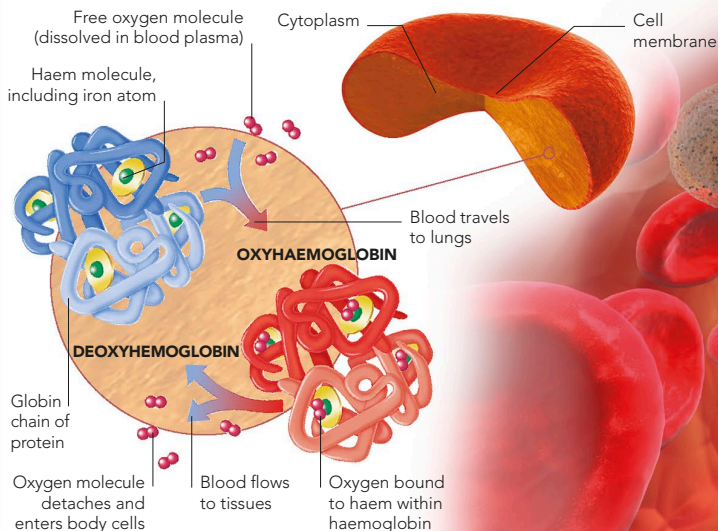
BLOOD AND BLOOD VESSELS

BLOOD IS A COLLECTION OF SPECIALIZED CELLS SUSPENDED IN A STRAW-COLOURED LIQUID CALLED PLASMA. FLOWING AROUND THE BODY, BLOOD CARRIES OXYGEN AND NUTRIENTS, COLLECTS WASTE, DISTRIBUTES HORMONES, AND SPREADS HEAT.

WHAT IS BLOOD?

An adult has about 5 litres (11 pints) of blood. Roughly 50–55 per cent of blood is plasma, 90 per cent of which is water. Plasma contains dissolved substances such as glucose (blood sugar), hormones, enzymes, and also waste products such as urea and lactic acid. Plasma also contains proteins such as albumins, fibrinogen (important in clotting), and globular proteins or globulins. Alpha and beta globulins help to transport lipids, which are fatty substances such as cholesterol.

Gamma globulins are mostly the disease-fighting substances known as antibodies. The remaining 45–50 per cent of blood is made up of three types of specialized cells. Red blood cells or erythrocytes carry oxygen; various white blood cells, known as leucocytes, are part of the defence system; and platelets or thrombocytes, which are tiny fragments of much larger cells, are involved in the process of clotting.



RED BLOOD CELLS AND HAEMOGLOBIN

A biconcave disc with no nucleus, each red blood cell contains 300 million haemoglobin molecules. Haemoglobin is composed of haem, an iron-rich pigment, and globin, ribbon-like protein chains. Oxygen in the lungs latches onto haem to make oxyhaemoglobin. In this form, oxygen travels through the bloodstream to all parts of the body.

BLOOD MAKE-UP

In 1mm³ ($\frac{1}{16,000}$ in³) of blood float about 5 million red cells, 10,000 white cells, and 300,000 platelets. These cells may have to move in single file through the narrowest blood vessels.